ODRA BIRA

AI/ML Engineer · LLM Applications & RAG · Generative Models · Computer Vision

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SUMMARY

M.S. Computer Science (AI/ML) candidate at the University at Buffalo (**3.83 GPA**) with **3 engineering roles** in deep-learning research and full-stack development. Ships LLM apps end to end (multi-provider FastAPI, hybrid RAG) and trains diffusion and vision models on GPUs.

TECHNICAL SKILLS

Languages: Python, Java, SQL, C/C++
Generative AI & LLMs: Claude & Gemini APIs, Ollama, vLLM, RAG, embeddings, vector search, BM25 hybrid retrieval, cross-encoder reranking, structured outputs, prompt engineering, LoRA/QLoRA fine-tuning (Hugging Face TRL, PEFT)
Machine Learning & CV: PyTorch, TensorFlow/Keras, scikit-learn, diffusion models, U-Net, YOLOv8, OpenCV, MediaPipe, transfer learning
Backend & Data: FastAPI, Flask, Pydantic, RESTful APIs, React, Node.js, PostgreSQL, MongoDB, Spark, Pandas, NumPy
MLOps & Tools: Git, Docker, Linux, CUDA GPU servers (SSH, tmux), pytest, Weights & Biases, Gradio, Streamlit

EXPERIENCE

University at Buffalo | *Graduate Student Researcher* *Oct 2025 – Present*

- Reproduce and benchmark safety-constrained diffusion planners (**SafeDiffuser, ICLR 2025**) as baselines for lab research on safe planning.
- Trained a **temporal U-Net diffusion trajectory planner** (600-step horizon, 1M iterations) and a value function for guided sampling on D4RL Walker2d on an **NVIDIA RTX 5090** GPU server.
- Benchmarked QP-based vs. closed-form **control barrier function (CBF)** enforcement over 100 episodes each: both met the safety constraint, and the closed form cut planning latency **3.2× (0.64 s → 0.20 s)**.

Accentiqa Systems Pvt. Ltd. | *Full Stack Developer* *Jun 2025 – Aug 2025*

- Delivered full-stack features for enterprise workflow apps, owning both the React front end and the RESTful back-end services.
- Designed relational database schemas and built end-to-end CRUD flows across UI, API and data layers.

Risposta Software India Pvt. Ltd. | *Software Developer Intern* *May 2024 – Aug 2024*

- Cut API data latency **20%** by profiling the slow SQL queries behind REST endpoints and rewriting them.
- Cut page-load time **30%** by reworking how React/Node.js features fetched and rendered data; built React UI and REST APIs.

PROJECTS

StudyEase – LLM-Powered Study Planner | *FastAPI, React, Pydantic, Claude/Gemini, Ollama, pytest* github.com/odra-bira/Studyplanner

- Architected a full-stack LLM app (React/Vite + async FastAPI, ~1.5K-line back end) with **5 typed REST endpoints** and PDF/image upload.
- Designed a **provider-agnostic LLM layer** (adapter pattern) over Claude, Gemini and local Ollama, switchable by config.
- Built a deterministic **validation-and-repair engine** that enforces date windows, daily capacity and 15–180 min session bounds on every LLM-generated plan, with rule-based fallback on model failure; **35 pytest tests** cover all layers.

SkinGPT – RAG Knowledge Assistant | *Embeddings, vector search, BM25, cross-encoder, LLM APIs*

- Engineered a **RAG pipeline** (chunking, embeddings, hybrid BM25 + dense retrieval, cross-encoder reranking) with cited answers.
- Built a **1,000-question evaluation benchmark** to measure retrieval quality and drive tuning, reaching **~75% retrieval precision**.

ShadowFix – Deep Learning Image Restoration | *PyTorch, U-Net, AMP, OpenCV, Gradio* github.com/odra-bira/ShadowFix

- Trained 3 U-Net variants (RGB, residual, mask-guided; ~7.8M params) on ISTD with mixed precision on an NVIDIA T4. Mask conditioning reached **30.19 dB PSNR / 0.951 SSIM** on 540 test images: **+3.0 dB** over the RGB baseline and +9.6 dB over the raw input.
- Split PSNR by region to show residual models over-correcting non-shadow pixels (−1.3 dB); deployed the best model as a Gradio app with automatic (OpenCV LAB-space mask) and paint-guided modes.

Trash Detection & Cleanliness Scoring | *YOLOv8n, OpenCV, Gradio, pytest* github.com/odra-bira/trash_detection

- Fine-tuned YOLOv8n on TACO to **0.78 precision**; added a 0–100 cleanliness score and density heatmap in a CLI + Gradio app.

Real-Time Pose Detection & Exercise Tracking | *MediaPipe, OpenCV, NumPy* github.com/odra-bira

- Built a real-time webcam pose tracker that computes shoulder–elbow–wrist joint angles from MediaPipe landmarks and counts reps with a two-state (160° / 30°) movement counter.

Pintos Operating System Kernel | *C, x86, QEMU* *Team of 3 · UB CSE 521*

- Extended the Pintos x86 kernel in C: alarm clock without busy-waiting, priority scheduling with nested **priority donation**, a 4.4BSD **MLFQS** scheduler with fixed-point math, **13 system calls**, user-stack argument passing and exec/wait synchronization.

EDUCATION

University at Buffalo, SUNY · M.S. Computer Science (AI/ML) · GPA 3.83/4.00 Aug 2025 – Dec 2026
Coursework: Machine Learning, Deep Learning, Data-Intensive Computing, Statistical Data Mining, Algorithms, Databases, OS, Security

Keshav Memorial Institute of Technology (JNTUH) · B.Tech. Computer Science (AI & ML) · GPA 8.4/10 Aug 2021 – May 2025
Coursework: NLP with Transformers, Deep Learning for Vision Systems, Big Data with Hadoop & Spark, Data Structures & Algorithms

PUBLICATION

Bira, O., et al. "Automated Assessment of Fruit Quality Using Transfer Learning and MobileNetV2." *IJRASET* 13(4), 2025.